Mainul Hossain

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Summary

Versatile and solutions-driven Site Reliability Engineer with a strong background in cloud infrastructure, automation, and systems administration. Extensive expertise in managing and optimizing Microsoft Azure and AWS environments, with hands-on experience in Docker, Kubernetes, and infrastructure-as-code practices. Skilled in scripting with Python, Bash, and Ruby for automation, monitoring, and API integrations.

Research interests include machine learning, deep learning, and explainable AI (XAI), with practical exposure to cybersecurity and system reliability. Adept at collaborating across teams, adapting quickly to evolving technologies, and driving operational excellence in both cloud and hybrid infrastructures.

Work Experience

Site Reliability Engineer - Optimizely Inc. Jun 2025 - Present

Part of **Reliability/Platform Engineering** Team

- Created AI-powered intelligent infrastructure monitoring tool and Agent Prompt Optimization that automatically analyzes webapplication performance for predicting Black Friday readiness with built-in PowerShell scripts for automated collection of performance data, used by AI to create actionable recommendations for smart server scaling
- Implemented a Terraform-managed Datadog Workflow that auto-restarts the Datadog agent on Elastic Search nodes via Azure RunCommand, with VM/VMSS detection, auditable success/failure events, and interval renotification escalation to oncall engineers.
- Contributed to maintaining the reliability, scalability, and performance of Optimizely's cloud-native SaaS infrastructure through proactive monitoring and observability implementations.
- Part of on-call and support SRE, resolving production issues and improving service reliability.
- Implemented well-scoped automation tasks within internal systems, including CI/CD enhancements and containerized workflow optimization, reducing manual operational overhead.
- Built and maintained comprehensive observability stack using Datadog, monitoring, logging, and tracing to ensure system reliability and performance across production environments.
- Collaborated across SRE, Operations, and Engineering teams to support service delivery, application lifecycle operations, and define SLOs/SLIs for critical services.
- Implemented Azure automation for .NET Core proactive monitoring and diagnostics, enhancing system observability and reducing incident response time.
- Continued to expand knowledge in system lifecycle management, infrastructure-as-code practices, and secure deployment workflows in AWS and Azure environments.

Managed Services Engineer - Optimizely Inc. Mar 2024 - June 2025

Part of **Strategic Support Solutions** Team

- Deployed containerized Python automation solutions using Docker/Kubernetes for auditing and cost optimization processes.
- Implemented infrastructure-as-code practices for Azure service optimization, reducing operational costs by 25%.
- Developed Python and Bash automation scripts for API integrations, data extraction, and troubleshooting workflows.
- Enhanced CI/CD pipelines using GitHub Actions for microservices deployment, reducing deployment time by 30%.
- Built monitoring and alerting solutions using Azure KQL and Datadog for distributed system observability.
- Administered Linux-based systems and managed AWS/Cloudflare CDN services for high-availability architectures.
- Maintained incident management processes and provided L2/L3 support for production system outages.

Associate Managed Services Engineer - Optimizely Inc. Sep 2022 - Mar 2024

Part of **Incident Management** Team

- Led incident response during high-volume traffic surges and major system outages, ensuring rapid detection and resolution.
- Managed production deployments and database changes with zero-downtime deployment strategies.
- Administered F5 load balancers and network security configurations for distributed systems.
- Served as Azure Administrator for DXP platform, maintaining 99.5% uptime across multiple regions.
- Implemented standardized incident management procedures, improving escalation processes and documentation.

Information Technology Officer - University Of Scholars Mar 2022 - Sep 2022

Part of **Information Technology** Team

- Designed and maintained Linux/Windows server infrastructure for university database and web systems.
- Deployed Windows Server domain controllers and implemented organization-wide monitoring solutions.
- Managed LAN/WAN network architecture and IP device configurations for campus connectivity.

Education

May 2015 - Jan 2022 B.Sc in Computer Science - **BRAC University** (GPA: 3.14/4.0)

Technical Skills

Programming Language -	Python, Ruby, JavaScript, Java, C, PHP
Cloud Platforms -	Microsoft Azure, AWS, Google Cloud Platform
Infrastructure-as-Code -	Terraform, Ansible, Azure DevOps
Scripting & Automation -	Python, Bash, PowerShell
Containerization -	Docker, Kubernetes, Container Orchestration
AI workflow -	Prompt Engineering, AI workflow automation, LLM integration, Agent architecture and tool design
Observability Tools -	Datadog, Azure KQL, ELK Stack (Elasticsearch, Logstash, Kibana), NewRelic, Pingdom
CI/CD Platforms -	GitHub Actions, Azure DevOps, Octopus Deploy
Operating Systems -	Linux (Ubuntu, CentOS, RHEL), Windows Server
Databases -	MySQL, Azure SQL Database, Amazon S3
Network & Security -	F5 Load Balancers, Cloudflare CDN, Network Configuration

Key Projects

Infrastructure Self-Healing Agent GitHub

An infrastructure focused auto-remediation bot that detects application failures, triggers automated recovery actions, and sends real-time incident notifications.

Pingdom API Automation & Monitoring GitHub

Python automation script for concurrent monitoring checks with CSV reporting for observability.

Cloudflare Infrastructure Automation GitHub

Dockerized Python solutions for DNS/WAF data extraction and infrastructure management.

.NET Core Monitoring & Diagnostics GitHub

Automated diagnostic tooling for Linux-based .NET Core environments with runbook integration.

Certifications & Awards

Microsoft Certified: Azure Administrator Associate Issued June 2024

Microsoft Corporation - **Credential ID MS0995597094**

Spot Award Issued Aug 2023

Optimizely Spot Award for "Never Stop Improving"

Spot Award Issued Apr 2023

Optimizely Spot Award for "Getting it done together"

Introduction to Programming Using Java Issued Mar 2018

LICT Project, ICT Division, Government of Bangladesh - **Credential ID G047514**